

Electrolysis

Electrolysis: The breakdown of an ionic compound to its molten or aqueous form by the passage of electricity.

Electrolytic cell: An electrochemical device that uses energy to facilitate a non-spontaneous redox reaction

Electrolyte: Aqueous solution or molten substance that undergoes electrolysis

Cathode: Negative electrode that attracts positive ions (cation) and where reduction takes place

Anode: Positive electrode that attracts negative ions (anions), and where oxidation takes place.

* Electrodes are made of carbon or platinum because they conduct electricity and are unreactive

* Electricity moves through ions in the internal circuit.

* Electrons move from the anode to the cathode in the external circuit.

Electrolysis: Molten Substance:

→ Metals are attracted to the cathode because they are positively charged. They get reduced.

→ Non-metals are attracted to the anode because they are negatively charged. They are oxidized.

Electrolysis: Aqueous solution

At the Cathode

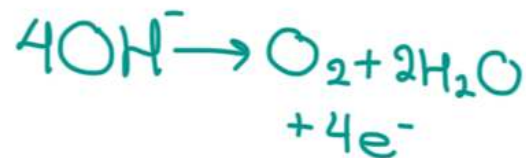
If the metal is less reactive than hydrogen, (Copper, Silver, gold) the metal will be reduced

If the metal is more reactive than hydrogen, hydrogen will be reduced



At the anode

No Halogens:
Hydroxide will be oxidized



Concentrated halogen solution.
halogen will be oxidized.
Diluted:
hydroxide will be oxidized

Differences between using graphite electrodes and copper electrodes:
(metal refining)

Electrolyte: Aqueous copper sulfate solution.

Graphite electrodes

At the cathode: Copper is reduced: $\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}$

At the anode: Hydroxide is oxidized: $4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4e^-$

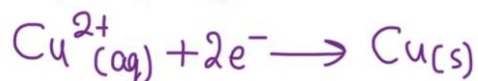
Observations:

- 1- Accumulation of copper at the cathode
- 2- Bubbles formation at the anode because oxygen is formed
- 3- The blue color of the solution fades because copper ions are reduced

Ions left in the solution: $\text{H}^+, \text{SO}_4^{2-} \rightarrow \text{H}_2\text{SO}_4$

Copper electrode: Pure copper cathode
Impure copper anode

At the cathode: Aqueous copper is reduced to solid copper



At the anode: Solid copper (electrode) is oxidized to aqueous copper



Observations:

- 1- Anode mass decreases as $\text{Cu}_{(\text{s})}$ is oxidized
- 2- Cathode mass increases as $\text{Cu}_{(\text{aq})}$ is reduced to $\text{Cu}_{(\text{s})}$, and accumulates at the cathode
- 3- The blue color of the solution will not fade, as reduced copper ions are replaced by the oxidized ions

Electroplating: Using electricity to coat one metal with another metal to make it look better or to prevent corrosion

* Electrolyte: Metal nitrate / Metal sulfate solution

* At the anode: The metal used to electroplate another object.

* At the cathode: The metal to be electroplated.